An LSTV-Stratified Audit of Production Spine CT Segmentation: Wrong-Level Surgical Risk and the Empirical Limits of Direct Semantic Vertebra Labeling

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Abstract

Wrong-level spine surgery—operating on the wrong vertebral level—remains a persistent never-event. Its single largest anatomic driver is the Lumbosacral Transitional Vertebra (LSTV), present in 5–35% of the population. We audit TotalSegmentator (henceforth the SOTA system), the de-facto production CT segmentation model, and show that on lumbarization cases its output pattern is anatomically indistinguishable from a wrong-level surgical plan: the segmenter lacks an L6 class and shifts its lumbar label assignment caudally by one anatomic level (per-class DSC L1=0.878, L4=0.388, L5=0.291), with junction Dice falling 29.0 points on Any-LSTV cases. This failure is invisible to every existing CT benchmark because none stratify by transitional anatomy. Recent work (VERIDAH [Möller et al., 2026]) addresses transitional vertebra labeling through a two-stage architecture: a per-vertebra classifier trained with merged L5/L6 (and T12/T13) labels, post-processed by a constrained sequence predictor. We make four contributions. (i) An LSTV-stratified evaluation protocol (per-class DSC, junction-DSC over a 40 mm L5/S1 window, per-class voxel confusion at the junction) that translates the level-shift mechanism into quantifiable wrong-level surgical risk. (ii) CT-SPINOPELVIC1K, a CT-native benchmark of 1,153 abdominal CT volumes across 802 patients with unified spinopelvic masks and 33 LSTV-positive cases across a 6-way phenotype taxonomy. (iii) The first LSTV-aware spinopelvic semantic segmenter—neither VERIDAH, SPINEPS, nor VerSe-derived models include the pelvis, and existing pelvic segmenters are not LSTV-aware. (iv) An empirical isolation experiment: adopting VERIDAH's training-side L5/L6 merge in the semantic-segmentation setting eliminates the L5/L6 collision that drives SOTA's level shift, but exposes a residual L4/last_lumbar collision on count-style sacralization cases that semantic segmentation alone cannot resolve. This residual collision empirically isolates the role of VERIDAH's sequence predictor as the necessary disambiguation component—a direction VERIDAH's authors could not test. CT-SPINOPELVIC1K, the protocol, trained checkpoints, and a 3D Slicer extension are released publicly.

1 Introduction

Wrong-level spine surgery is a persistent never-event in operative spine care [Konin and Walz, 2010]. Despite intraoperative imaging and surgical-pause checklists, the most common cause—failure to correctly count vertebral levels in the presence of an LSTV—continues to account for the majority of

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reported wrong-level lumbar incidents [Carrino et al., 2011]. The mechanism is well documented: a surgeon counts lumbar bodies upward from the sacrum, finds five (because lumbarization presents a sixth lumbar body that has not been recognized), and labels the bottom body as L5 when it is anatomically L6. Surgery proceeds one level caudal to the intended target.

Modern radiology workflows increasingly route automated segmentation as an upstream substrate for clinical decision-making. TotalSegmentator (henceforth the SOTA system, the audit subject) [Wasserthal et al., 2023] is the de-facto production model: an nnU-Net trained on 1,204 routine CTs to delineate 104 anatomic structures, with thousands of clinical and research installations. The original paper reports a test-set Dice of 0.943 across all classes and qualitatively documents two known spine failure modes: "mixing neighboring vertebrae classes" (1% of cases) and "mixing neighboring rib classes" (12%). This paper concerns one anatomic class on which the SOTA system does not perform at its headline level, and the consequences when it does not.

The clinical claim. On lumbarized spines, the SOTA system produces an output pattern that is, when consumed by surgical planning pipelines or by surgeons confirming operative levels, *indistinguishable from a wrong-level surgical plan*. Its label vocabulary contains L1–L5 but no L6 class. On a lumbarized spine (six lumbar bodies plus sacrum), the model distributes its five lumbar labels across the six bodies by shifting caudally by one anatomic level: the body that is anatomically L6 receives the L5 label, the body that is anatomically L5 receives the L4 label, and so on. A surgeon targeting the disc between predicted-L4 and predicted-L5 lands on the anatomical L5–L6 disc, one level caudal to the intended L4–L5 target. the SOTA system reproduces the canonical LSTV-induced miscount automatically and silently.

The architectural alternative: VERIDAH. Recent work [Möller et al., 2026] addresses transitional vertebra labeling with a two-stage architecture. Stage one is a 3D DenseNet169 classifier on per-vertebra crops with four classification heads: vertebra label (23 classes), region (cervical/thoracic/lumbar), vertebra transition (six classes including "last lumbar"), and visibility. Stage two is a constrained sequence predictor that finds the optimal label sequence subject to anatomical-count constraints (4–6 lumbar; allow $L5 \rightarrow L5$ once, interpreted as L6). The classifier is trained with L5/L6 labels *merged*: "Since especially T13 cases are rare, training directly on class labels like T13 would create a huge class imbalance problem for the classifier. To circumvent this, we replace the T13 and L6 labels in the reference annotation with T12 and L5" [Möller et al., 2026, §2.2]. The merge is a class-imbalance device; disambiguation between L5/L6 (equivalently, between an anatomical L4 in a 4-vertebra spine and L5 in a 5-vertebra spine) is deferred to the sequence predictor's anatomical-count constraints, not solved by the classifier. The architectural commitment has been validated empirically by VERIDAH (97.22% subject-level CT agreement) but has not, to our knowledge, been tested against the alternative of direct semantic segmentation: does a sufficiently expressive voxel-wise network, trained at full resolution with explicit LSTV stratification and the same training-side merge, require the sequence predictor at all? This paper closes that gap.

The evaluation gap. the SOTA system does not report LSTV-stratified performance. Neither does the VerSe benchmark [Sekuboyina et al., 2021], CTSpine1K [Deng et al., 2021], or CTPelvic1K [Liu et al., 2021]. Earlier CT vertebra labelers explicitly excluded the failure mode [Sekuboyina et al., 2020]: "we only considered 24 labels for C1–L5 and did not account for segmentation anomalies, such as L6, or transitional vertebrae, such as a lumbarized S1 vertebra." VERIDAH reports performance on subjects containing thoracolumbar and lumbar enumeration anomalies but on a private cohort and without isolating semantic-segmentation behavior from sequence-predictor contribution. The result is a measurement vacuum: a production model that emits a confident, locally well-formed, caudally-shifted prediction on a 5–35%-prevalence congenital variant produces no error signal at evaluation time.

Closing the architectural gap requires hardware. Demonstrating that the gap is closable requires a baseline that incorporates LSTV cases in training *and* fits sufficient anatomical context to learn the L5/S1 spinopelvic relationship. At spacing $0.75 \times 0.75 \times 0.80$ mm and patch $256 \times 320 \times 320$ (the only patch size fitting both the lumbar spine and pelvis on resampled COLONOG geometry), a single nnU-Net training batch consumes $\sim$100 GB of GPU memory under `nnUNetResEncUNetPlans_100G`. This budget is available on H200 hardware (140 GB HBM3e) but not on the L40S/A100/H100 GPUs that bracket the SOTA system's reported training configuration. Neither VERIDAH nor SPINEPS

train at this combined spinopelvic resolution: SPINEPS operates on the spine alone, and VERIDAH's per-vertebra classifier sees only a local crop, not the lumbosacral context.

Contributions.

1. **Wrong-level surgical risk in SOTA.** Systematic one-level caudal shift on lumbarization quantified through per-class DSC collapse (L4=0.388, L5=0.291), per-class voxel confusion at L5/S1 (95%+ of GT-L6 voxels labeled L5 on representative cases, §4), and structural absence of L6 prediction across all 1,152 evaluation cases.
2. **CTSpinoPelvic1K: a stratified spinopelvic benchmark.** 1,153 abdominal CT records across 802 patients, unified spinopelvic masks, 33 LSTV-positive patients across a 6-way subtype taxonomy (lumb $n = 14$; sacr_count $n = 9$; full sacralization $n = 6$; semi-sacralization $n = 2$; ambiguous $n = 2$), and a registration-free fusion pipeline that resolves a previously undocumented Y-axis coordinate-convention mismatch between CTSpine1K and CTPelvic1K. Both source datasets share the COLONOG cohort that underlies VerSe and CTSpine1K, but neither encodes pelvic context with LSTV stratification.
3. **First LSTV-aware spinopelvic semantic segmenter.** A 5-fold CV nnU-Net 3d_fullres baseline on H200 hardware adopting VERIDAH's training-side merge (last_lumbar = L5+L6) at the voxel level, combined with pelvic-class supervision in a single forward pass. VERIDAH and SPINEPS operate on the spine alone, and CTPelvic1K-trained pelvic segmenters are not LSTV-aware.
4. **Empirical isolation: merge alone is insufficient.** The merge eliminates the L5/L6 collision driving SOTA's level shift but exposes a residual L4/last_lumbar collision on count-style sacralization. This residual collision empirically isolates the component of VERIDAH's success attributable to its sequence predictor—a direction VERIDAH's authors could not ablate.

2 Related Work

Wrong-level spine surgery and LSTV. Konin and Walz [2010] review LSTV-induced miscounting; Carrino et al. [2011] quantify the effect of segment variants on radiologist-assigned levels; Sekharappa et al. [2014] report prevalence between 5% and 35% depending on modality and classification. None of these works addresses automated segmentation as a propagation pathway for the same miscount.

VERIDAH and two-stage vertebra labeling. Möller et al. [2026] report 97.22% subject-level CT agreement (vs. 77.26% for the prior CT state-of-the-art [Meng et al., 2023]) using a per-vertebra DenseNet169 classifier with four heads, trained with L5/L6 (and T12/T13) labels merged for class-imbalance reasons. Disambiguation is deferred to a constrained sequence predictor that imposes anatomical-count constraints and explicitly allows one $L5 \rightarrow L5$ transition (interpreted as L6). VERIDAH's ablation (their Table 5) shows the four-head + sequence-predictor configuration outperforms a single-head classifier or classifier-only-with-sequence-predictor, but the comparison is never made against direct semantic segmentation. Our CTSpinoPelvic1K-trained segmenter (§7) constitutes that missing comparison.

SPINEPS and CT vertebra upstream. SPINEPS [Möller et al., 2024] is a whole-spine T2w MRI segmenter and is the upstream that VERIDAH uses on the MRI side. VERIDAH's CT path was trained with manually-annotated instance masks; for new-CT deployment the VERIDAH authors point to VerSe-derived [Sekuboyina et al., 2021] models as upstream segmenters. Both SPINEPS and VerSe-derived upstream segmenters are spine-only, which is the gap our spinopelvic segmenter closes.

CT spine and pelvis benchmarks. Sekuboyina et al. [2021] introduce VerSe, a challenge-format CT spine benchmark of 460 volumes that encodes transitional anatomy via its label vocabulary (label 25 = L6) but does not stratify by LSTV subtype and is not paired with pelvic ground truth. The earlier 2D-reformation labeling work [Sekuboyina et al., 2020] explicitly excluded LSTV cases. CTSpine1K [Deng et al., 2021] (1,005 CT volumes with VerSe-derived per-vertebra annotation) and CTPelvic1K [Liu et al., 2021] (1,184 pelvic CTs including the COLONOG sub-cohort) draw from the COLONOG cohort but are unstratified by LSTV subtype.

Table 1: SOTA zero-shot stratified by LSTV phenotype on CTSPINOPELVIC1K, full-resolution mode. Junction DSC over a 40 mm L5/S1 axial window. L6 count: cases with any L6 voxel emitted (structurally zero by label vocabulary). †: junction analysis not computed for separate-mode records.

Subgroup	n	**L1**	**L2**	**L3**	**L4**	**L5**	**Sac**	**Hips**	**Jxn DSC**
All cases	801	0.911	0.906	0.913	0.914	0.913	0.817	0.929	0.818
Normal	768	0.921	0.918	0.929	0.936	0.938	0.817	0.929	0.824
Any LSTV	33	0.681	0.615	0.523	0.367	0.314	0.822	0.929	0.534
Lumbarization	14	0.878	0.853	0.690	0.388	0.291	0.833	0.928	**0.357**
Sacralization (count-style)	9	0.265	0.100	0.055	0.001	0.000	0.814	0.926	0.697
Semi-sacralization	2	0.951	0.951	0.951	0.955	0.956	0.824	0.937	†
Sacralization (full)	6	0.718	0.676	0.673	0.724	0.793	0.826	0.932	0.636
Ambiguous	2	0.952	0.951	0.882	0.678	0.374	0.771	0.932	†
L6 EMIT									0 / 1152

Multi-source label fusion and modality-adjacent LSTV detection. Li et al. [2025] combine CTSpine1K and CTPelvic1K but treat them as independent sources without resolving the coordinate-convention mismatch we describe in §5. Kwak et al. [2025] detect LSTV on plain radiographs (binary task, different modality). The position that benchmarks should report stratified safety-critical performance [Maier-Hein et al., 2024] motivates the protocol of §6.

3 Failure-Mode Analysis: SOTA on LSTV

We evaluate the SOTA system v2 (`total` task) zero-shot at full resolution on the full CTSPINOPELVIC1K matched cohort. the SOTA system was not trained on any CTSPINOPELVIC1K case, so every case is valid; this provides N=1,152 successful records aggregated to 801 patients via patient-level merging, including 33 LSTV-positive patients. Full-resolution mode isolates the failure as a property of label vocabulary and training distribution rather than inference resolution.

Table 1 reports performance by phenotype. On normal anatomy ($n = 768$) the SOTA system performs at its published level. On Any-LSTV ($n = 33$) every metric drops; lumbarization collapses caudally (L1=0.878 $\rightarrow$ L4=0.388 $\rightarrow$ L5=0.291) with junction-DSC at 0.357. Count-style sacralization ($n = 9$) collapses uniformly across all lumbar classes (L1=0.265 down to L5=0.000). L6 predictions are structurally zero across all 1,152 cases.

The level-shift mechanism. The asymmetry between lumbarization (severe, caudally progressive) and count-style sacralization (uniform collapse) follows from a single fact: the SOTA system's output head has no L6 class. For 6-vertebra anatomies the model compresses 6 anatomic units into 5 labels by shifting label assignment caudally; the most rostral body remains L1 (escape into thoracic vertebrae), but every successive body is misaligned, with the final body labeled L5 when it is anatomically L6. For 4-vertebra anatomies the model emits standard L1–L5, overshooting the GT lumbar count and producing a $+1$-level offset uniformly. The level-shift mechanism is the segmentation analog of the surgical miscount [Konin and Walz, 2010].

4 Case Study: Wrong-Level Surgical Risk in SOTA Output

The aggregate metrics establish the population pattern. The case study below establishes the mechanism at the level a surgeon or planning system would consume the output: per-class voxel confusion at the L5/S1 junction with explicit translation into the surgical-target error. Token 167 is a fused-mode lumbarization case with junction-DSC 0.221, the lowest among lumbarization cases. Table 2 shows the 40 mm-window confusion matrix: 92,537 of 92,629 GT-L5 voxels (99.9%) are predicted as L4, and 81,088 of 85,186 GT-L6 voxels (95.2%) are predicted as L5—the level-shift mechanism in its cleanest form.

Surgical translation. A surgeon planning an L4–L5 microdiscectomy and using the segmentation to confirm operative level identifies the disc between predicted-L4 (which sits on anatomical L5) and predicted-L5 (which sits on anatomical L6). The targeted disc is the L5–L6 disc, one level

Table 2: Token 167 (lumbarization) per-class voxel confusion matrix over the 40 mm L5/S1 window. Bold cells: dominant prediction class for each GT row. The diagonal-shifted dominance pattern (GT-L5 → pred L4; GT-L6 → pred L5) is the empirical signature of the level-shift failure.

GT \ Pred	bg	L3	L4	L5	sacrum	L6
L5	4,572	2	**92,537**	90	0	0 (n/a)
L6	4,089	0	9	**81,088**	0	0 (n/a)
sacrum	68,635	0	0	36	**80,334**	0 (n/a)

Table 3: CTSPINOPELVIC1K composition. LSTV counts are patient-level. Per-record subtype refinement (§5) gates each record's training-time LSTV tag on whether the defining anatomy is supervised in that record's labels.

Property	Value	LSTV cases
Patients matched (CTSPINOPELVIC1K)	802	—
Records matched (CTSPINOPELVIC1K)	1,153	33 patients (44 records)
Fused (spine + pelvis)	342	—
Separate (spine + pelvic)	352	—
Spine-only / Pelvic-only	90 / 20	—
Lumbarization	—	14
Sacralization (count-style)	—	9
Sacralization (full)	—	6
Semi-sacralization	—	2
Ambiguous	—	2

caudal to the L4–L5 intent. The output is locally well-formed: five compact vertebral predictions in roughly correct positions. The error is detectable only by counting bodies (six lumbar, not five), which production downstream pipelines do not perform.

5 CTSPINOPELVIC1K: Construction and Composition

CTSPINOPELVIC1K fuses CT scans from COLONOG [Smith et al., 2015] with annotations from CTSpine1K [Deng et al., 2021] (VerSe-derived per-vertebra masks) and CTPelvic1K [Liu et al., 2021] (sacrum and bilateral hips). The same COLONOG cohort underlies VerSe and CTSpine1K; CTSPINOPELVIC1K's contribution is the patient-level fusion of CTSpine1K's vertebra masks with CTPelvic1K's pelvic masks plus the LSTV-stratification overlay neither source provides. Of 1,153 matched records, 342 carry fused (spine + pelvis) ground truth, 352 are separate-mode, 90 are spine-only, and 20 are pelvic-only (Table 3).

Patient-anchored fusion and coordinate correction. Patient identity is resolved via canonical UID matching (825/825 patients). Each mask is assigned to its source CT acquisition by intra-patient bone-HU maximization, $s^*(m) = \arg\max_{s \in \mathcal{S}_p} |\{v \in \mathcal{L}(m,s) : I_s(v) > 200\,\text{HU}\}|/|\mathcal{L}(m,s)|$, which serves as a metadata-free oracle since `SeriesDescription` fields are unreliable in TCIA releases (99.6% spine, 100% pelvic placement). CTSpine1K and CTPelvic1K were converted from DICOM with different `dcm2niix` invocations, producing a systematic Y-axis sign reflection in NIfTI affines; we detect via affine-derived center sign and reverse along axis 1 with affine update.

Unified label map. We adopt VERIDAH's training-side merged-label scheme at the voxel level: 0 = bg, 1–4 = L1–L4, **5 = last_lumbar** (anatomical L5 in 5-vertebra spines; both L5 and L6 in 6-vertebra spines), 6 = sacrum, 7–8 = left/right hip, 9 = ignore.

Six-way LSTV taxonomy with dual-provenance labels. We retain both LSTV label sources separately and partition into six subtypes by combining filename-qualifier evidence (CTPelvic1K morphological criterion) with vertebral-counting evidence (CTSpine1K VerSe-label-presence heuristic): **normal**, **lumb** (6 vertebrae labeled or VerSe label 25 present), **sacr_count** (4 vertebrae labeled with sacralization flag), **semisacralization** (filename qualifier), **sacralization** (filename qualifier), and **ambiguous** (cross-source disagreement). Earlier extractions collapsed semisacralization into

full sacralization due to a substring-match parser bug; we identify and fix this. Without the 6-way taxonomy, the qualitatively different failure patterns of full sacralization, semi-sacralization, and count-style sacralization (§3) would be invisible.

Partial-annotation training protocol. Training only on fused records discards 70% of the data. We use nnU-Net v2's `ignore_label` [Isensee et al., 2024] to train on all 1,153: voxels in unsupervised regions are written as label 9, contributing zero to Dice and cross-entropy. Spine classes (1–5) receive supervision from 784 records; pelvic classes (6–8) from 714. This is a $3.36\times$ effective expansion. Per-record subtype refinement gates training-time tags on supervised anatomy.

6 Evaluation Protocol

We define four metrics as the minimum required to attribute LSTV failure modes to wrong-level surgical risk [Maier-Hein et al., 2024]: (1) per-class DSC stratified by 6-way LSTV subgroup, exposing class-level degradation; (2) junction-DSC over a 40 mm L5/S1 axial window, the clinical window for surgical planning; (3) per-class voxel confusion at the L5/S1 junction, which exposes structured failure modes (the level shift) that DSC alone cannot characterize; and (4) for the merged-label baseline, three focused per-epoch confusion blocks anchored on the L4/last_lumbar boundary that drives the residual collision (`last_lumbar_GT_on_lumb`, `L4_GT_on_sacr_count`, `pred_last_lumbar_on_sacr_count`) plus a specificity number (`last_lumbar_specificity_on_sacr_count`).

7 Spinopelvic Baseline

Architecture. nnU-Net v2 [Isensee et al., 2020, 2024] ResEnc-M planner, configuration `3d_fullres`, plans `nnUNetResEncUNetPlans_100G` (100 GB GPU-memory target). Voxel spacing $0.80\times0.75\times0.75$ mm, patch $\mathbf{256\times320\times320}$, batch size 2. Encoder: 7-stage residual encoder with $n_{\text{features}}=[32, 64, 128, 256, 320, 320, 320]$, $n_{\text{blocks}}=[1, 3, 4, 6, 6, 6, 6]$, $3\times3\times3$ kernels, $2\times2\times2$ strides at every stage transition, instance normalization, in-place LeakyReLU, deep supervision at five resolution levels. Symmetric decoder with $n_{\text{conv_per_stage}}=1$. Output is 9 channels (background + 8 foreground); ignore-label is masked from loss rather than emitted as a head.

Loss and optimization. Class-reweighted DC_and_CE loss with deep-supervision weights $[0.516, 0.258, 0.129, 0.065, 0.032]$ and per-class CE weights $\mathbf{w}=[0.5, 1, 1, 1, 1, \mathbf{1.5}, \mathbf{2.0}, 1, 1]$ for [bg, L1, L2, L3, L4, last_lumbar, sacrum, left_hip, right_hip]; the last_lumbar boost ($1.5\times$) targets the merged class as the clinically-important boundary, sacrum ($2\times$) supports Castellvi-distinguishing precision, background (0.5) suppresses dominant-class gradient. SGD with Nesterov momentum 0.99, weight decay 3×10^{-5}, PolyLR from 10^{-2} over 500 epochs (250 train + 50 val iterations per epoch).

Two-tier LSTV emphasis. Queue-level oversampling. All five non-normal subtypes are duplicated in the dataloader's keys list until they constitute 25% of training samples per fold, matching the $4\times$ duplication rate VERIDAH applies to anomalous cases. **Class-reweighted CE** (above). We do not apply per-patch foreground-class biasing: under merged labels, biasing toward an L6 class is ill-defined, and biasing the L4/last_lumbar boundary on sacr_count cases would confound the residual-collision measurement.

Hardware and runtime. H200 NVL (140 GB HBM3e), 24 CPU workers, preunpacking once per job to a shared local-NVMe cache (1.7 TB `/tmp`, $\sim$3 GB/s, replacing NFS reads that throttled to 21 MB/s under 5-fold parallel contention). Per-epoch wall-clock $\approx$ 400–700 s; per-fold ≈ 3 days for 500 epochs. Reducing GPU memory budget forces either a smaller patch (loses spinopelvic spatial relationship) or coarser spacing (loses sub-millimeter Castellvi-defining boundary detail); the 100 GB budget is the smallest fitting both constraints. 5-fold patient-stratified CV with 6-way LSTV stratification.

Table 4: CTSPINOPELVIC1K 5-fold CV baseline on held-out validation folds, 6-way taxonomy. Δ vs. SOTA zero-shot (Table 1). LL EMIT: cases where the model emitted ≥ 100 last_lumbar voxels. [Placeholders fill at training completion.]

Subgroup	n	**L1–L4**	**last_lumbar**	**Sac**	**Hips**	**Jxn DSC**	**LL emit**
Normal	XX	0.XXX	0.XXX	0.XXX	0.XXX	0.XXX	XX/XX
Lumbarization	14	0.XXX	**0.XXX**	0.XXX	0.XXX	**0.XXX**	XX/14
Sacralization (count)	9	0.XXX	—	0.XXX	0.XXX	0.XXX	X/9
Semi-sacralization	2	0.XXX	0.XXX	0.XXX	0.XXX	0.XXX	X/2
Sacralization (full)	6	0.XXX	0.XXX	0.XXX	0.XXX	0.XXX	X/6
Ambiguous	2	0.XXX	0.XXX	0.XXX	0.XXX	0.XXX	X/2
Δ vs. SOTA (lumb)	—	+XX.X	N/A	+X.X	0	**+XX.X**	—
Δ vs. SOTA (sacr_count)	—	+XX.X	—	+X.X	0	+X.X	—

Table 5: Residual class confusion under merged labels, focused on L4/last_lumbar. Block 1: GT-last_lumbar on lumb (good signal). Block 2: GT-L4 on sacr_count (failure-mode signal). Block 3: pred-last_lumbar on sacr_count (inverse direction). [Numbers fill at training completion.]

Block	last_lumbar	sacrum	L4	background
Block 1: last_lumbar GT on lumb	**XX.X%**	X.X%	X.X%	X.X%
Block 2: L4 GT on sacr_count	*X.X%*	X.X%	**XX.X%**	X.X%
Block 3: pred-last_lumbar on sacr_count	—	X.X%	**XX.X%**	X.X%

8 Results

Result tables populated from completed 5-fold CV.

Headline. The merged-label spinopelvic baseline closes the LSTV junction-DSC gap on lumbarization by $\Delta =$ **[+XX.X]** points (SOTA = 0.357 $\rightarrow$ ours = **[0.XXX]**, mean across 5 folds) and emits non-degenerate last_lumbar on **[XX/14]** lumbarization cases with mean per-case last_lumbar DSC = **[0.XXX]**, eliminating the structural absence that drives SOTA's level shift. A residual class collision persists on count-style sacralization (per-case L4 DSC = **[0.XXX]**); §8.1 characterizes this as the empirical signature of direct semantic segmentation's limit on transitional anatomy.

8.1 The Residual Class Collision

The merged scheme eliminates the L5/L6 collision driving SOTA's wrong-level mechanism: GT-last_lumbar voxels on lumb cases are predicted as last_lumbar in **[XX]%** of voxels (Block 1), against SOTA's structurally-zero last_lumbar emission. But it exposes a residual collision at the L4/last_lumbar boundary on sacr_count cases: GT-L4 voxels are predicted as last_lumbar in **[XX]%** of voxels (Block 2), and predicted-last_lumbar voxels on sacr_count cases are anatomically L4 in **[XX]%** (Block 3). Specificity = **[0.XXX]**.

Interpretation. The model has learned the spinopelvic spatial relationship on normal anatomy and recovers the merged last_lumbar class on lumbarization. But on count-style sacralization the model's positional prior dominates: "the bottom-most lumbar visible in this patch is last_lumbar" regardless of the global count. This is the same reasoning failure that VERIDAH's per-vertebra classifier head, by design, cannot solve in isolation: the disambiguating signal (vertebra count from a known anchor) is non-local. A $256 \times 320 \times 320$ patch (or, in VERIDAH's formulation, a per-vertebra crop) does not reliably contain both the caudal lumbosacral and rostral cervico-thoracic boundaries needed to count.

This is a finding rather than a defect of the baseline. It executes the cleanest test of VERIDAH's training-side merge in the semantic-segmentation setting and confirms the architectural hypothesis underlying VERIDAH's two-stage design: the merge addresses class imbalance, but disambiguation between "last lumbar in a 4-vertebra spine" and "last lumbar in a 5-vertebra spine" must be deferred to a downstream component with access to global anatomical structure.

9 Discussion and Future Work

Generalization and hardware. The level-shift mislabeling failure is a property of any spine segmentation model whose output head excludes L6, deployed on a distribution containing lumbarization. The 100 GB patch budget is presently available only on H200-class hardware; readers who cannot replicate it should expect a quantifiably worse LSTV gap closure and therefore residual wrong-level risk in any downstream pipeline that consumes the output.

Future Work. (1) **VerSe expansion via sacrum pseudolabeling.** Direct semantic segmentation may yet resolve the residual collision given more LSTV-stratified training data. VerSe [Sekuboyina et al., 2021] contains 460 CT volumes and explicitly labels L6 (label 25) but lacks pelvic ground truth. Our segmenter's sacral DSC (**[0.XX]** on held-out folds) is competitive with the best CTPelvic1K-trained baseline and exceeds SOTA's sacral DSC (0.817), making it a viable sacrum pseudolabel source for VerSe. The union enables a $1{,}613$-volume LSTV-stratified spinopelvic training cohort for testing whether scale alone resolves the L4/last_lumbar collision without the merge-and-count pipeline. (2) **Two-stage integration with VERIDAH.** Our spinopelvic segmenter can replace VERIDAH's VerSe-derived CT upstream by supplying the sacrum boundary directly (VERIDAH's "last lumbar" transition anchor) and higher-resolution per-vertebra centroids in a single forward pass. (3) **Scaling hypothesis.** The residual collision reflects, we conjecture, the limits of a 33-patient LSTV-positive training cohort rather than a fundamental architectural limit. With ≥ 200 LSTV-positive cases (achievable through VerSe, the VERIDAH-reported NAKO sub-population, and emerging public benchmarks), direct semantic segmentation on a 9-class *unmerged* scheme with explicit L6 may close the collision the merge currently masks—rendering the two-stage pipeline an artifact of data limitations rather than an inherent commitment. (4) **3D Slicer deployment.** We package the segmenter as a 3D Slicer extension as a drop-in replacement for the spinopelvic regions of SOTA's output, the current Slicer default.

Limitations. COLONOG is a colonoscopy-screening cohort (US, ages 45–75); generalization to emergency or pediatric populations is not evaluated. LSTV labels are inherited from CTPelvic1K and CTSpine1K heuristics. Two subtypes (semi-sacralization, ambiguous) have $n = 2$ patients each. SOTA is evaluated zero-shot, the correct comparison for "SOTA as production system" but undersells what it could achieve if retrained with an L6 class on LSTV-stratified data—a follow-up enabled by CTSPINOPELVIC1K.

Implications. A production CT segmentation system that emits a confident, locally well-formed, caudally-shifted prediction on a 5–35%-prevalence congenital variant produces no error signal at inference time and, when consumed downstream, reproduces the canonical mechanism of wrong-level spine surgery. Any future spine segmentation model claiming readiness for surgical-planning integration should report per-subgroup DSC and per-class confusion at the L5/S1 junction on a stratified LSTV cohort.

Data and Code Availability

CTSPINOPELVIC1K is hosted on HuggingFace Datasets (anonymous repository for review) with both core and Responsible AI fields of the Croissant specification [Akhtar et al., 2024]; a 50-record sample covering all six LSTV subtypes is provided. The evaluation protocol, data-construction pipeline, and trainer source are released under permissive license at an anonymous GitHub repository accessible without credentialed access. Trained 5-fold checkpoints and a 3D Slicer extension will be made available.

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